

Exercise Sheet 6

November 21th: OO-Principles I

Exercise 1

We investigate the Use Case “The Payroll Case Study” in Robert Martin’s book

1. We will find the right abstractions looking at the specification and use cases! Read the pages from the Martin’s book (see below). Also the documents `Principles_and_Patterns.pdf` and `PayrollCaseStudy.pdf` might be helpful!
2. We will apply design patterns and OO-principles learned so far!
3. We will analyze and criticize our design!

Tasks:

1. Preparation: You must read the pages 147-151 and p 192-250 in *Robert Cecil Martin. Agile Software Development: Principles, Patterns, and Practices. Prentice Hall PTR, Upper Saddle River, NJ, USA, 2003* or alternatively, pages 293-298 and p 349-363 in *Robert C. Martin and Micah Martin. Agile Principles, Patterns, and Practices in C# (Robert C. Martin). Prentice Hall PTR, Upper Saddle River, NJ, USA, 2006*.
2. Working in pairs, prepare a short summary of your understanding of the design (UML)!
3. Think about alternative designs!
4. Plenum discussion to verify understanding, clarify open points and exchange ideas and assessment

Exercise 2

Consider an application for an online book store with the following class:

```
public class Book {
    //...
    String title;
    String author;
    int isbn;

    public void buy() throws Exception {
        if (checkAvailabilityFromInventory()) {
            invoice();
            ship();
        } else
            throw new NotAvailableException();
    }

    public boolean checkAvailabilityFromInventory() {
        if (oracleDB == true)
            return loadFromOracleDb(this);
        else
            return loadFromMySqlDb(this);
    }

    public void invoice() {
        // send invoice to customer
        // ...
    }
}
```

```
    }  
  
    public void ship() {  
        // ship book to customer  
        // ...  
    }  
}
```

1. List all violations of SOLID principles you find!
2. Change (i.e. refactor) the code to make compliant to SOLID principles¹! Use patterns you know already and list them!

Exercise 3

In the context of the previous exercise assume you get the change request to include ebooks, too. Therefore you define a class like

```
public class EBook extends Book{  
    public void ship() {  
        // ship book to customer  
        enableDownloadLink(this);  
    }  
}
```

1. Does this design obey the LSP?
2. What do you think of the implementation of the ship method?
3. Suppose you include an attribute **weight** in **Book** to enable calculation of shipping costs. A developer in your team sets **weight** to zero for ebooks. What do you think about this design decision?
4. How would you implement support for ebooks?

Exercise 4

SOLID has been critized by software developers, for instance in <https://speakerdeck.com/tastapod/why-every-element-of-solid-is-wrong> or <https://qualityisspeed.blogspot.com/2014/08/why-i-dont-teach-solid.html>, but also has been defended by many, not just its “inventor” Robert C. Martin, see e.g. <https://blog.ndepend.com/defense-solid-principles/>.

1. Write down some main arguments against SOLID (as a list)!
2. What do you think of his arguments against SOLID?

Hints

- Consult the literature!
- You can work in pairs, if you want!
- If you want to learn a Java API, look into the java docs!
- Always use the same familiar IDE (suggestion JetBrains IntelliJ or Eclipse)!

¹use dummy print statements to log what should happen